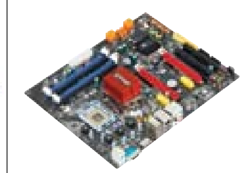
Motherboards	Boards Up To Rs 10,000			
Brand Model No.	MSI K9A2 Neo2	MSI P43 Neo	ECS P45T-AD3	Gigabyte EP43-DS3L
				
Chipset	AMD RX780	Intel P43 Neo	Intel P45	Intel P43
Socket	AM2/AM2+	LGA 775	LGA 775	LGA 775
Northbridge / Southbridge	AMD RX780 / AMD SB700/SB750	P43 / ICH10	P45 / ICH10R	P43 / ICH10
Price (Rupees)	4500	5200	7560	8500
Plus (+)				
Minus (-)				
Grand Total (Out of 100)	54.01	57.57	61.89	64.47
Features (out of 60)	24.36	22.50	26.10	29.22
Performance (Out of 40)	29.65	35.07	35.79	35.25
Front Side Bus Speed (MHz)	2000 MHz	1333 MHz	1333 MHz	1333 MHz
Memory Type	DDR2	DDR2	DDR3	DDR2
Max Memory Supported (GB) / No of DIMMs	8 GB / 4	8 GB / 4	8 GB / 4	8 GB / 4
No. of SATA / IDE Ports	6 / 1	6 / 1	6 / 1	6 / 1
E-SATA Support / No Of E-SATA Ports	✘ / NA	✘ / NA	y / 1	✘ / NA
No of PCI Express Graphics Slots	1	1	2	1
No. of PCI / PCI x1 / PCI x4 Slots	3 / 2 / ✘	3 / 2 / ✘	2 / 2 / ✘	2 / 4 / ✘
Multi GPU Support (SLI/CrossFire/N) / No of PCIe Lanes for multi GPU setups	N / NA	N / NA	Y / Crossfire x8	✘ / NA
Integrated Audio Controller	Realtek ALC888	Realtek ALC 888	Realtek ALC 883	Realtek ALC 888
SPDIF / Optical Port	✘ / ✘	✘ / ✘	✘ / ✘	✔ / ✔
No of LAN Ports / 10/100 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit
Integrated Wi-Fi (Y/N)	✘	✘	✘	✘
No. of rear USB / FireWire Ports	6 / 0	4 / ✘	6 / 0	8 / 1
All Solid State Caps (Y/N)	✘	✘	✘	✔
BIOS options				
Overclocking / Overvolting (So 10)	5 / 5	5 / 5	3.5 / 3.5	5.5 / 5.5
CPU voltage range (volts)	(-)26.6 to (+)30.4%	-0.4 ~ +0.7875v	+50mV ~ +328mV	0.5 ~ 2.0v
Memory voltage range (volts)	1.80 to 2.90v	1.416 ~ 3.324v	1.5v ~ 2.2v	1.8 ~ 2.5v
Package Bundle				
No of SATA / PATA Cables Provided	4 / 1	3 / 1	4 / 1	3 / 1
CD's	Drivers	Driver, Backup software	Drivers	Drivers
Other Miscellaneous Accessories	NA	NA	USB and FireWire bracket	NA
Performance				
PC Mark 2005:				
CPU / Memory	7623 / 4496	9563 / 6433	9683 / 6691	9553 / 6326
Graphics / HDD	16450 / 6837	18253 / 6527	18429 / 7017	18425 / 6922
Overall	8774	10138	10573	10467
SiSoft Sandra 2008:				
CPU Arithmetic (Dhrystone ALU / Whetstone iSSE3)	39365 / 33387	54667 / 44028	55530 / 44066	55343 / 44016
CPU Multimedia (Integer / Floating)	98644 / 130006	328936 / 180293	329473 / 180768	329272 / 180264
Drive Index (MB/s) / Random Access Time (ms)	71 / 6	73 / 6	72 / 6	72 / 6
SiSoft Memory bandwidth score (Integer / Floating)	5309 / 5301	6526 / 6253	7196 / 7199	6526 / 6527
3D Mark 2006 Marks / CPU Score	13415 / 3850	16638 / 4949	16879 / 4975	16727 / 4962
CINEBENCH R10 (CPU)	8319	11824	11833	11828
WinRAR 3.8	1844	2192	2263	2193
Video Encoding (DivX 6.8) Time (sec)	80	114.3	114.2	115.4
File Transfer Test (4 GB File) (MBps)	145.2	136.5	136.7	135.2
Doom 3 FPS (1600x1200, 4xAA, 4xAF / 640 x 480, low detail)	172.7 / 185.5	203.2 / 288.4	212.7 / 305.4	207.6 / 301.3
S.T.A.L.K.E.R Shadow Of Chernobyl (1680x1050, 4xAA, 4xAF / 800 x 600, low detail)	136.2 / 443.19	180.3 / 894.6	180.95 / 916.11	180.23 / 888.4
Prey (1680 x 1050, 4x AA, 4x AF, 640 x 480, low detail)	165.8 / 227	204.3 / 348.8	211.2 / 357	202.2 / 336.84

it warrants mention. The board itself is good looking with an all-black PCB and is the best laid out board in this category. The EP43-DS3L from Gigabyte was cleanly laid out and was the only one to sport an all solid-state capacitor design. We like the placement of the memory slots and the SATA ports in particular. The CPU region was also fairly un-cramped. One issue was the Northbridge cooler isn't fixed on too tight and uses plastic push-through pins to fasten to the board. Due to this the cooler was quite loose, which isn't good for heat dissipation. This board was the only one to

have eight rear USB ports for hassle free connectivity. MSI's P43 Neo with its fire engine red PCB is clearly a basic solution as MSI uses a black PCB for their high-end boards. It has a simple aluminium heatsink on the Northbridge and is cleanly laid out with a lot of gap between the PCIe slot and PCI slots.

MSI P45 Platinum

Performance

The P45 chipset is slightly faster than the P43 and ECS' P45T-AD3 takes the lead in this regard but by the finest of margins. Of course this could also be attributable to the use of DDR3 memory especially in case of the memory bandwidth benchmarks which are largely theoretical in nature. The MSI K9A2 Neo2 is just eaten up here; and not for any fault of its own. AMD really trails behind Intel in terms of processor performance and this reflects in most of the benchmark scores.

Our Winner(s)

All in all, MSI's P43 Neo impressed the most. It trailed the fastest board by a hairsbreadth and was a whole Rs. 2,360 cheaper. At Rs. 5,200 its your best bet for a cheap yet powerful board that's reasonably feature-rich. It won our *Digit Best Buy* award in the sub-Rs. 10,000 category and represents superb value for money.

Motherboards above Rs. 10,000 and up to Rs. 18,500

Enthusiastically yours

The number of people willing to spend Rs. 10,000 or above on a motherboard are few; however this number is slowly on the rise. Overclockers, gamers demanding multi-GPU systems and others wanting a really feature rich board have no other choice. When you pay above Rs. 10,000 for a board you are not just looking for connectivity options or even mundane features.

You want bragging rights and flaunt value as well as rock solid performance and build quality. In addition these boards usually have better quality or higher grade components. This helps with increased life and stability while overclocking which is important for those interested in benchmarking and posting scores on forums and such. Some of these boards will also have fancy looking heatpipe cooling solutions; this is great for people having cabinets with transparent side windows or anyone wanting to show his components off.

Features

The boards in the category were all quite feature rich and diverse in terms of the chipsets they were based on. We found Intel's P45 boards aplenty along with a few of the cheaper X48 chipset based boards. NVIDIA's nForce 780i SLI chipset also makes an appearance here. A sole AMD offering based on the new 790FX was available from ASUS. ASUS themselves had five boards in this category. The attractively packaged P5Q Premium and P5Q3 Deluxe caught our eye first. Both are based on the P45 chipset; but the number three as a suffix on the P5Q3 Deluxe means this board supports DDR3 instead of the de facto DDR2. This board also supports two WLAN access points — nifty but useless for most people. The P5Q Premium actually has four RJ45 ports; a waste if you ask us, although we did like the 10 USB ports. Both these boards also support *Express Gate*, an ASUS patent for a separate Linux based firmware OS built on to a flash chip which allows you to surf the internet and use Skype without booting into Windows. Both boards have great looking heatpipe solutions and are very well laidout except for the fact that one SATA port remains unusable with full length graphics cards. This is because while the other SATA ports point outwards this one is vertically aligned (meaning it points upwards); connecting a hard drive to this port with a full length graphics card plugged in is impossible. We liked the fact that ASUS chose to supply one PCI slot above the primary PCIe slot. This means that you can use your PCI soundcard, even if you have a large custom cooler on your graphics card, without fear of airflow getting impeded in any way. Incidentally these boards support a 16-phase power system, good for ensuring an ultra stable supply of power to your CPU which is of importance to overclockers. Gigabyte's GA-P45-DS4P is another solidly built offering. It's a great looker too with a good looking heatpipe solution that doesn't take up too much space. Biostar's T-Power I45 was also a good looking and well laid out board and an add-on fan with heatpipe is also provided. Biostar uses funky colours for the expansion slots and the SATA ports are well placed; not so the PATA port which is a little awkward although not inaccessible.

ASUS' Striker Formula (780i chipset) and Rampage Formula (X48 chipset) are part of their built for gamers ROG (Republic Of Gamers) series. These boards have add-on soundcards featuring the same ADI 1988B audio chipset that ASUS uses across all their high-end motherboards. They also have small LED-backlit buttons for functions like power on/off, CMOS reset and system reset. This is convenient for people working with open systems (like us!) or overclockers. They also have LEDs on the PCB that indicate the current status of Northbridge, FSB and CPU overclocks. Then there is the huge yet unobtrusive one-piece heatpipe cooling solutions that ASUS uses; these cooling solu-